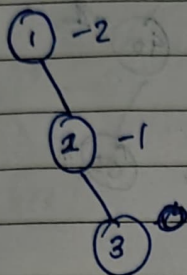
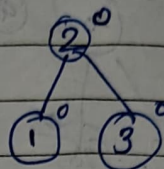


AVL Tree

1. Insert into an empty AVL tree:- 1, 2, 3, 4, 5, 6, 7. Name every rotation and draw the final tree.

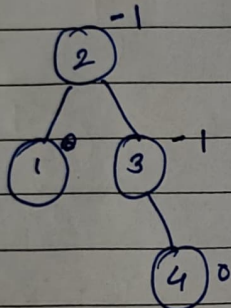


Left Rotate(1)



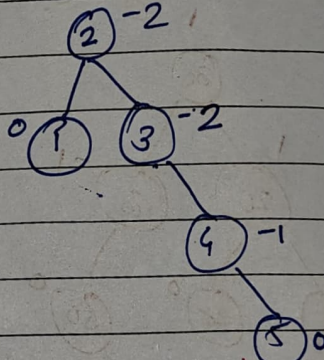
This is RR case at node 1 - RR rotation

* Insert(4)



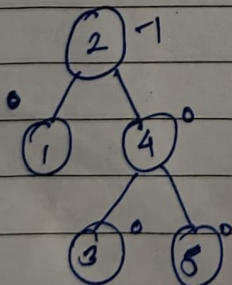
No rotation tree is Balanced

Insert(5)



first Unbalanced node = 3
This is RR at 3.

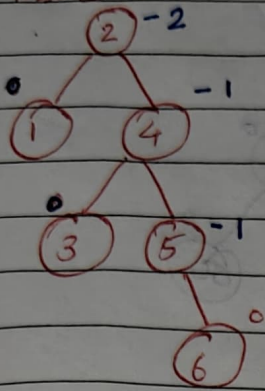
After Rotation (Left Rotate(3))



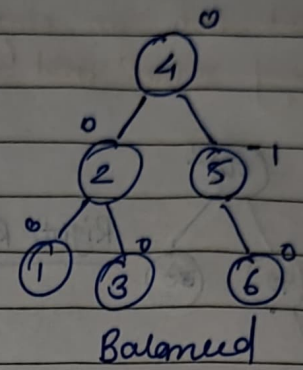
Now it is Balanced

★

Insert 6:



Left Rotate (2)



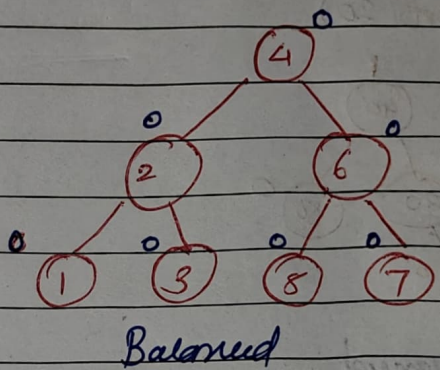
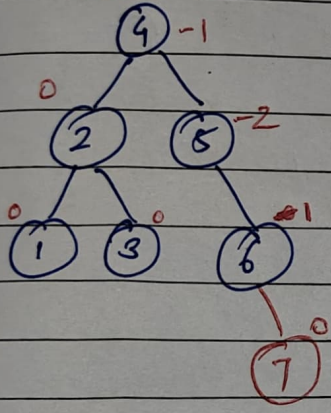
Balanced

Unbalanced node = 2.

This is RR case at 2.

★

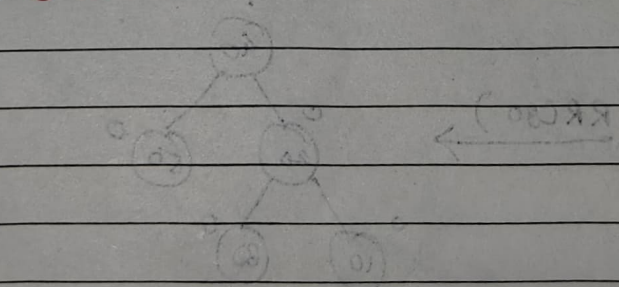
Insert 7:



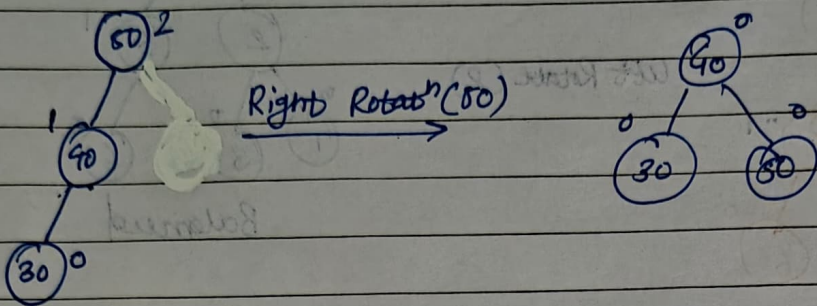
Balanced

unbalanced node = 5

This is RR at 5

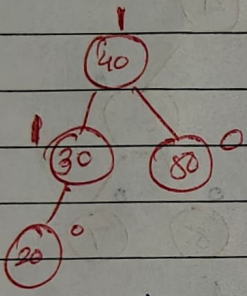


2. Insert into an empty AVL tree: 80, 40, 30, 20, 10. Name every rotation and draw final tree.



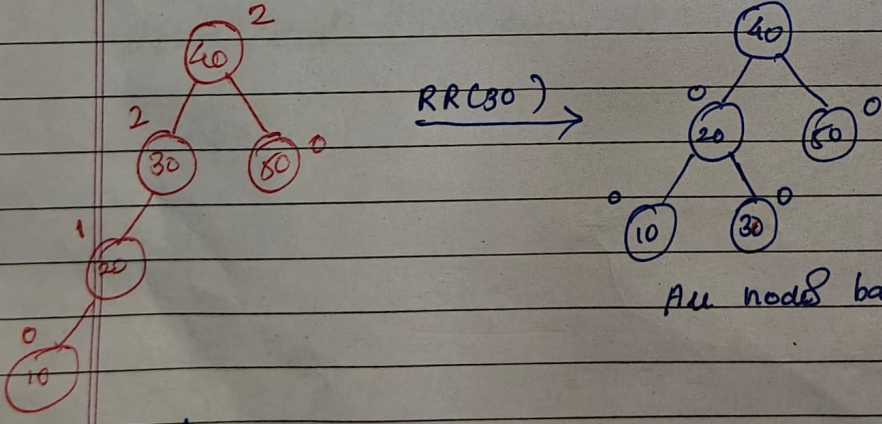
LL rotation Case
mbalanced

* Insert (20)



Balanced

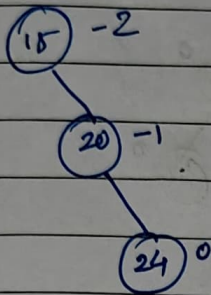
* Insert (10)



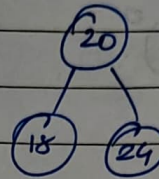
All nodes balanced

mbalanced
LL case

3. Insert into an empty AVL tree: 18, 20, 24, 10, 13, 7, 30, 36, 28
Name every rotation and draw the final tree.



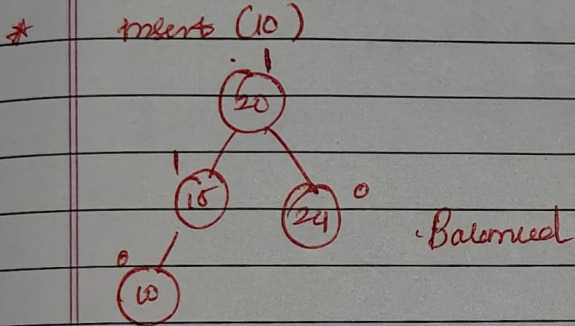
LR Rotatⁿ(18)



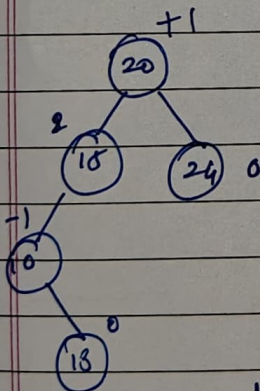
Balanced

Not Balanced

RR Case



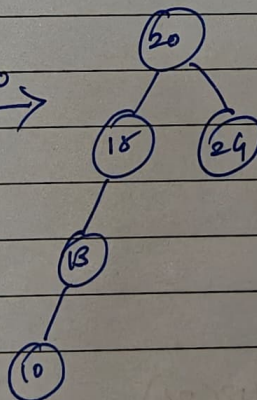
- * insert(13)



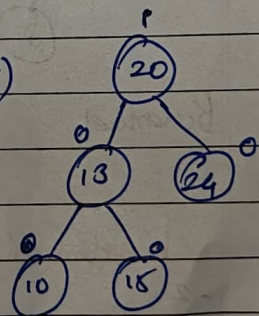
Unbalanced

LR Case

LR C10

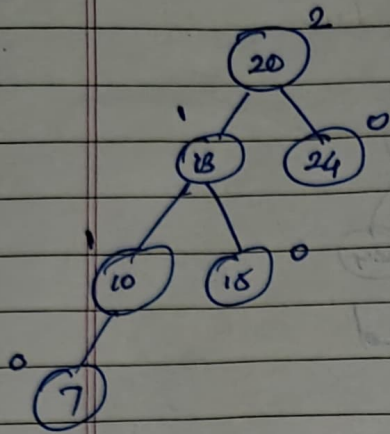


RR(18)



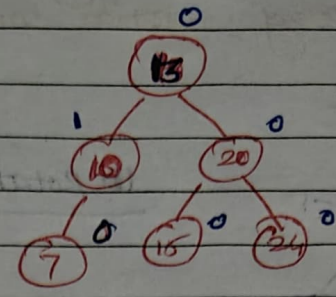
Balanced

* Insert (7)



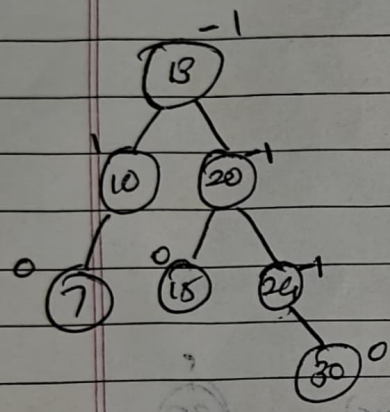
Node 20 is unbalanced
LL Case

RR(20) →



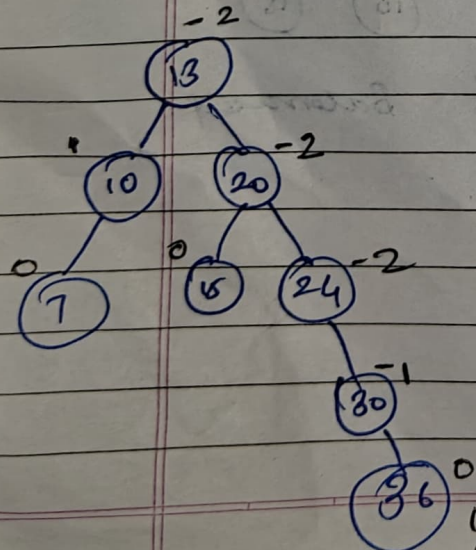
Balanced.

* Insert (30)



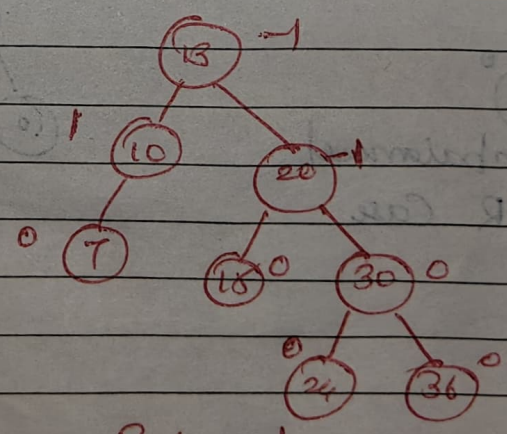
Balanced

* Insert (36)



Unbalanced
RR Case

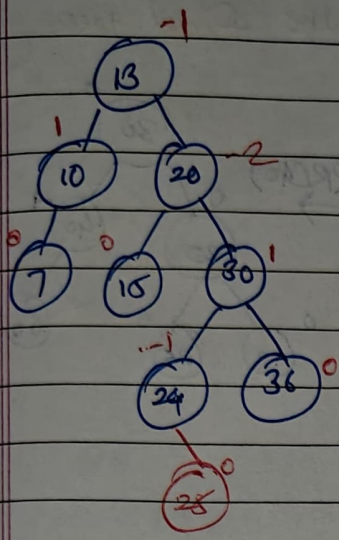
LRC(24) →



Balanced

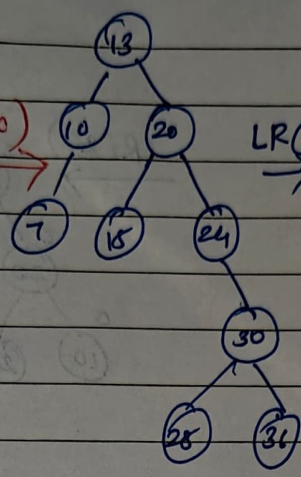
*

Insert (25)

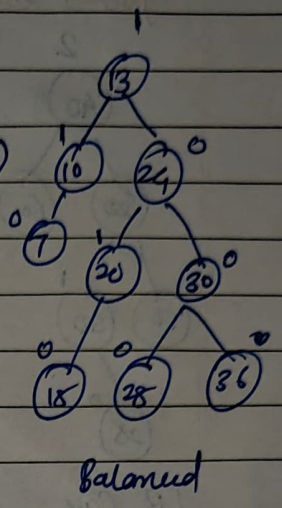


Unbalanced
RL case

RR(30)

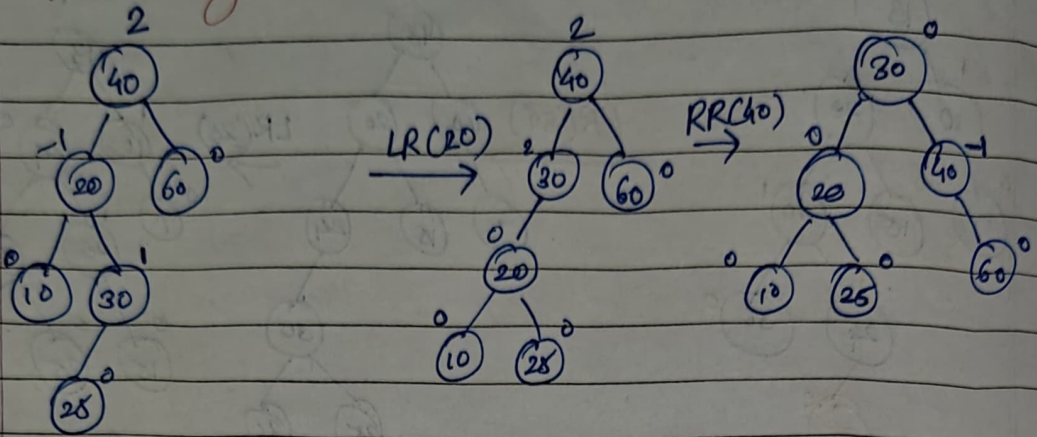


LR(20)



Balanced

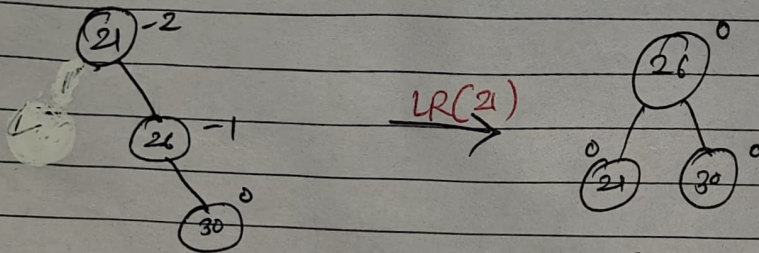
4. Insert into an empty AVL tree: 40, 20, 60, 10, 30, 28.
Name every rotation and draw the final tree.



LR Case

unbalanced

* Insert into an empty AVL tree: 21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3, 7. Name every rotation and draw Final tree.

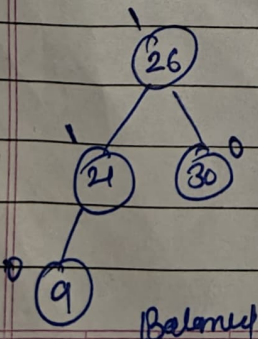


Balanced

unbalanced

RR case

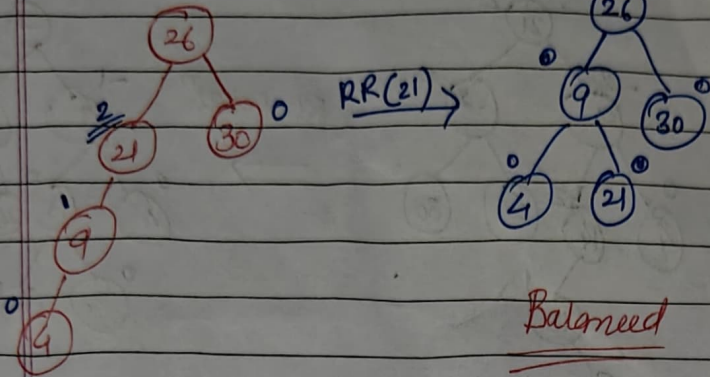
* insert(9)



Balanced

*

insert(4)

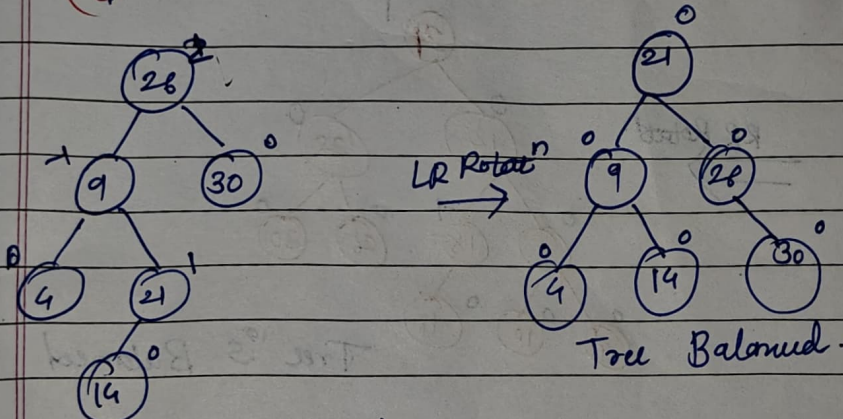


LL Case

unbalanced Node 21.

*

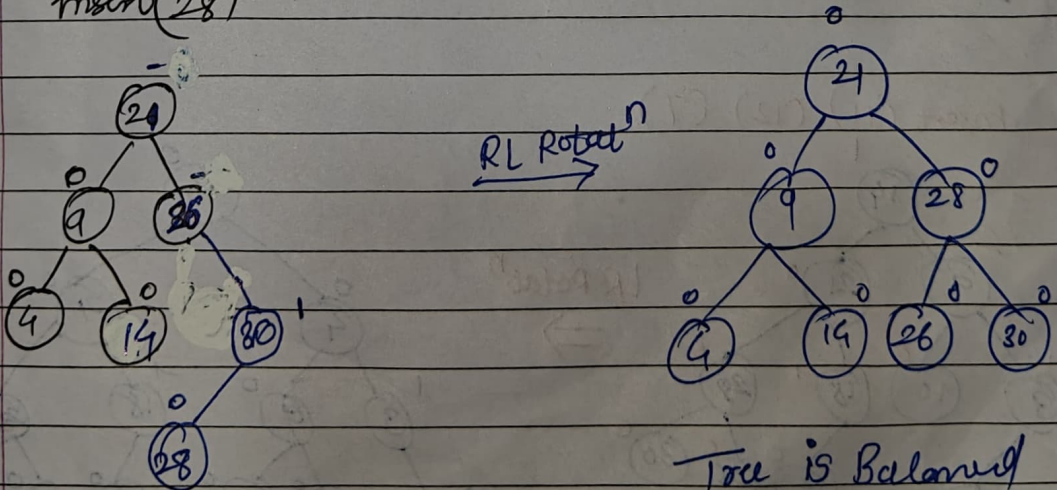
(14) insert



Tree is not balanced

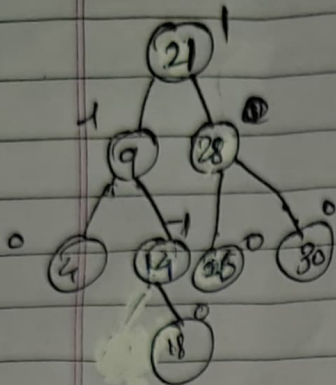
*

insert(28)



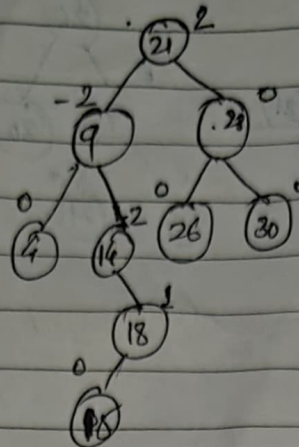
Tree is Not Balanced

* Insert (18) :



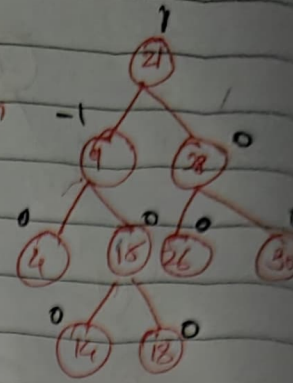
Tree Balanced

* Insert (15) :

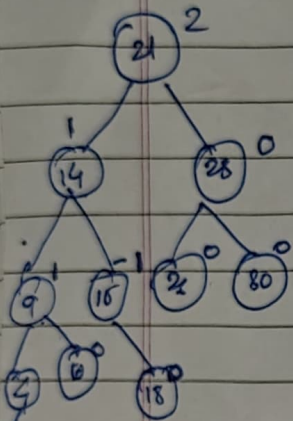


Tree is Not Balanced for 9 & 21

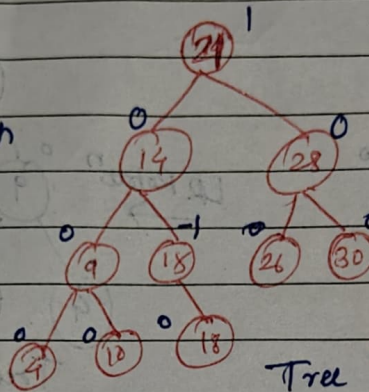
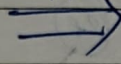
RL Rotatⁿ



* Insert (10) :



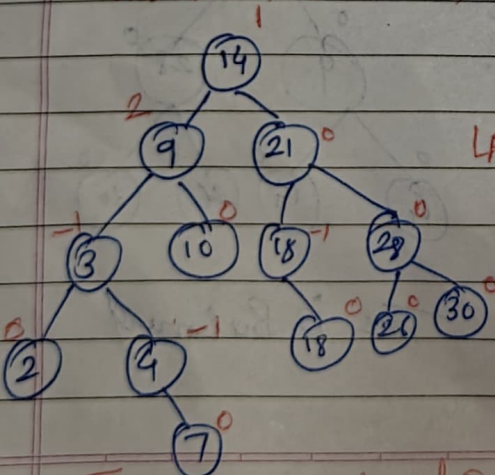
RR Rotatⁿ



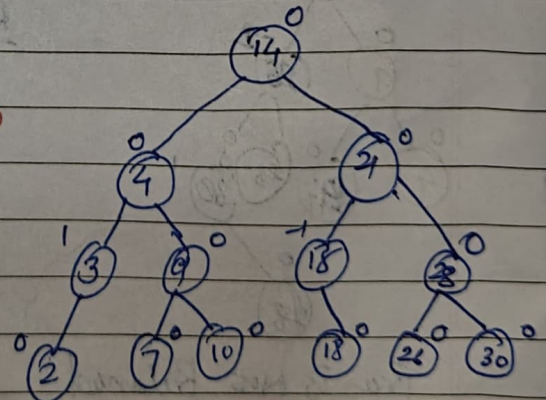
Tree is Balanced

Tree is Not Balanced for 21

* Insert (11) (12) (7) :



LR Rotatⁿ



Tree is Not Balanced for 9